

Brandon H. Schlomann

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Summary

- Biophysicist studying the dynamics of infections and immune responses
- Emphasis on model organisms: fruit flies and zebrafish
- Primary tools: live imaging, genetics, theory and computation

Positions

University of California, Berkeley: James S. McDonnell Postdoctoral Fellow

Berkeley, CA
2020-present

- Advisor: Hernan Garcia
- Discovered spatial compartments in a key fly tissue with distinct immune response dynamics
- Developed technology for live imaging of single-cell transcriptional dynamics in zebrafish

Education

University of Oregon: PhD in Physics

Eugene, OR
2014-2020

- Advisor: Raghuvier Parthasarathy
- Pioneered the study of gut bacterial dynamics in zebrafish via live imaging
- Discovered a mechanism by which sublethal concentrations of antibiotics can destabilize gut bacterial communities

University of Montana: BA in Physics, with Honors (minor in Mathematics)

Missoula, MT
2009-2013

- Advisor: Andrew Ware
- Designed the shape of superconducting coils used to confine fusion plasmas via large-scale numerical optimization

Awards and Fellowships

NIH Loan Repayment Program Award (NIAID)	2025
James S. McDonnell Foundation Postdoctoral Fellowship	2020–2023
Weiser Doctoral Thesis Award	2020
NIH Molecular Biology and Biophysics Training Grant	2016–2019
NSF GRFP Honorable Mention	2016
First Year Academic Award, Institute of Theoretical Science, U. Oregon	2014
Cal-Murphy Scholarship, U. Montana	2009

Publications

12 papers total, 8 first-author research papers, including 1 single-author paper

First-Author Research Papers (* = equal contributors)

Brandon H. Schlomann, William S. DeWitt, Yuanzhao Zhang, Kasturi Shah. *Ignition criteria for trigger waves in cell signaling* 🔗. arXiv (2025) doi: 10.48550/arXiv.2508.16810

Brandon H. Schlomann, Ting-Wei Pai, Jazmin Sandhu, Genesis Ferrer Imbert, Thomas G. W. Graham, Hernan G. Garcia. *Spatial microenvironments tune immune response dynamics in the Drosophila larval fat body* 🔗. bioRxiv (2024) doi: 10.1101/2024.09.12.612587

Elizabeth Eck*, Bruno Moretti*, **Brandon H. Schlomann***, Jordão Bragantini, Merlin Lange, Xiang Zhao, Shruthi VijayKumar, Guillaume Valentin, Cristina Loureiro, Daniele Soroldoni, Loïc A. Royer, Andrew C. Oates, Hernan G. Garcia. [Single-cell transcriptional dynamics in a living vertebrate](#) . bioRxiv (2024)
doi: 10.1101/2024.01.03.574108

Brandon H. Schlomann, Raghuvier Parthasarathy. [Gut bacterial aggregates as living gels](#) . eLife (2021) 10:e71105.
Press: : [Journal Club for Condensed Matter Physics](#)

Travis J. Wiles*, **Brandon H. Schlomann***, Elena S. Wall, Raghuvier Parthasarathy, Karen Guillemin. [Swimming motility of a gut bacterial symbiont promotes resistance to intestinal expulsion and enhances inflammation](#) . PLoS Biology, 18(3): e3000661

Brandon H. Schlomann*, Travis J. Wiles*, Elena S. Wall, Karen Guillemin, Raghuvier Parthasarathy. [Sublethal antibiotics collapse gut bacterial populations by enhancing aggregation and expulsion](#) . Proceedings of the National Academy of Science (2019), 166, 43, 21392-21400.

Press: [Nature Reviews Microbiology](#), [In Brief](#) , [Nature Reviews Microbiology](#), [Under the Lens](#) , [Trends in Microbiology](#)

Brandon H. Schlomann*, Travis J. Wiles*, Elena S. Wall, Karen Guillemin, Raghuvier Parthasarathy. [Bacterial cohesion predicts spatial distribution in the zebrafish intestine](#) . Biophysical Journal (2018), 115, 22712277

Brandon H. Schlomann. [Stationary moments, diffusion limits, and extinction times for logistic growth with random catastrophes](#) . Journal of Theoretical Biology (2018), 454, 154-163

Other Research Papers

Jarrett F. Lebov, **Brandon H. Schlomann**, Catherine D. Robinson, Brendan J.M. Bohannon. [Phenotypic parallelism during experimental adaptation of a free-living bacterium to the zebrafish gut](#) . mBio (2020) 11, 4, e01519-20

Travis J. Wiles, Elena S. Wall, **Brandon H. Schlomann**, Edouard A. Hay, Raghuvier Parthasarathy, Karen Guillemin. [Modernized tools for streamlined genetic manipulation and comparative study of wild and diverse proteobacterial lineages](#) . mBio (2018), 9, 5, e01877-18

Travis J Wiles, Matthew Jemielita, Ryan P Baker, **Brandon H Schlomann**, Savannah L Logan, Julia Ganz, Ellie Melancon, Judith S Eisen, Karen Guillemin, Raghuvier Parthasarathy. [Host gut motility promotes competitive exclusion within a model intestinal microbiota](#) . PLoS Biology (2016), 14(7): e1002517

Review Articles

Brandon H. Schlomann and Raghuvier Parthasarathy. [Timescales of gut microbiome dynamics](#) . Current Opinion in Microbiology (2019) 50:56–63

Workshops Organized

09/2025	3rd Physics of Life Symposium , Chan-Zuckerberg Biohub	San Francisco, CA
04/2024	Modeling the Evolutionary Dynamics of Cheating in Natural Viral Infections , Santa Fe Institute	Santa Fe, NM
10/2023	2nd Physics of Life Symposium , Chan-Zuckerberg Biohub	San Francisco, CA
10/2023	Slow-Fast Dynamics in Biological Systems , Santa Fe Institute	Santa Fe, NM

Conference Talks and Seminars

03/2025	Spatial microenvironments tune immune response dynamics in the Drosophila larval fat body , American Physical Society Global Physics Summit	Anaheim, CA
03/2024	Deciphering spatial patterns of immune response in fruit fly larvae , American Physical Society March Meeting	Minneapolis, MN
01/2023	Coarse-graining immune responses , Chan-Zuckerberg Biohub, Physics of Life Symposium	San Francisco, CA
10/2020	Antibiotics and aggregation in the zebrafish gut , Transport in the digestive tract: experiments, modeling, applications to microbiology	virtual

09/2020	<i>Kinetics and statistics of gut bacterial cluster sizes</i> , University of British Columbia, SBME Seminar	virtual
09/2019	<i>Live imaging of gut bacterial communities in zebrafish: spatial structure, fast dynamics, and antibiotic perturbations</i> , 2nd International Fish-Microbiota Workshop	Eugene, OR
08/2019	<i>Live imaging of gut bacterial communities in zebrafish: spatial structure, fast dynamics, and antibiotic perturbations</i> , PacNow qBio Meeting	Portland, OR
03/2019	<i>Bacterial cohesion predicts spatial distribution in the zebrafish intestine</i> , American Physical Society March Meeting	Boston, MA
03/2018	<i>Swim/stick tradeoffs for gut bacterial symbionts</i> , American Physical Society March Meeting	Los Angeles, CA
06/2017	<i>Catastrophe and fast dispersal dynamics of the zebrafish gut microbiota following sublethal antibiotics</i> , International Fish-Microbiota Workshop	Trondheim, Norway
03/2017	<i>Visualizing the response of a gut bacterial population to antibiotic perturbations</i> , American Physical Society March Meeting	New Orleans, LA
06/2016	<i>Response dynamics of gut bacterial communities</i> , META Center Symposium	Eugene, OR

Students mentored

2025	Ruby Knudsen , UC Berkeley Physics Undergraduate Student Project: theoretical studies of virus and immune cell spread in tissues.
2025	Jianjie Xu , UC Berkeley Physics PhD Student Rotation project: theoretical studies of transcriptional oscillators.
2024	Kensington Hartman , UC Berkeley Molecular and Cell Biology PhD Student Rotation project: theoretical studies of coupled transcriptional oscillators.
2023	Hana Mir , UC Berkeley Physics PhD Student Rotation project: theoretical studies of nonlinear wave generation in immune signaling.
2023-24	Jazmin Sandhu , UC Berkeley Physics Undergraduate Student Project: experimental studies of fruit fly immune responses, as part of the PI2 program.
2022-25.	Tiffany (Ting-Wei) Pai , UC Berkeley Molecular and Cell Biology Undergraduate Student Project: experimental studies of fruit fly immune responses, as part of the SURF program. Now a lab technician at UC Berkeley.
2022	Genesís Ferrer Imbert , UC Berkeley Biophysics PhD Student Rotation project: experimental studies of fruit fly immune responses and computational image analysis.
2021-22	Sathvik Anantakrishnan , IIT Madras Biological Sciences Masters student. Remote project: theoretical studies of coupled transcriptional oscillators. Now a Biophysics PhD Student at UC Berkeley.

Teaching, Outreach, DEI

2023-25	UC Berkeley Quantitative Biology Bootcamp. Each summer I led experimental rotations for an 8 day summer course introducing incoming biology graduate students to quantitative biology.
2023	UCSF TRAIN UP Inclusive Mentoring Course. I attended this course on lab management and mentorship.
2023-24	UC Berkeley POWER (Physical Sciences Opportunities for Womxn in Education and Research). I mentored two community college students and helped them prepare for transferring to a 4-year university.
2014-18	Summer Academy to Inspire Learning, University of Oregon. Each summer I led small group demonstrations for high school students during an annual summer outreach program.
2014-18	PHYS 204/205/206, University of Oregon. I taught one year of algebra-based physics lab.
2007-09	Physics Study Jam, University of Montana. I worked at a drop-in tutoring center, tutoring undergraduate students in various physics courses.